



△ **Brilliant** oval-cut, faceted moldavite

Moldavite

Moldavites were formed 15 million years ago, when a meteorite struck near modern Ries, Bavaria, melting the local sandstone. The substance is a tektite, the name for minerals formed when large meteorites hit Earth: the terrestrial rock melts on impact, is splashed into the air, and quickly cools to form the glass-like substance. Tektites are found on most continents, but moldavite is local to the Ries impact point. It is typically olive-green to dull greenish yellow, and is found in sizes ranging from less than a millimetre to several centimetres across.

Specification

Chemical name	Silicon dioxide	Formula	$\text{SiO}_2(+\text{Al}_2\text{O}_3)$
Colour	Mossy green, greenish yellow	Structure	Amorphous
Hardness	5.5	SG	2.40
		RI	1.48–1.54
		Lustre	Vitreous
		Streak	n/a
		Locations	Germany, Czech Republic



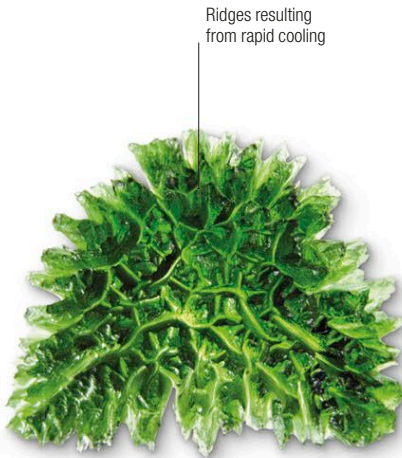
Irregular surface

Spindle shape | Rough | During the flight of molten moldavite through the air, it assumes a number of shapes, such as this elongated form, called a spindle.



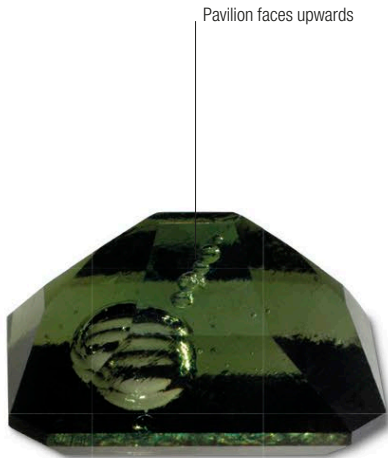
Coloured layers

Uncut moldavite | Rough | The substantial, chunky form of this moldavite rough suggests that it cooled extremely rapidly after being thrown from the meteorite crater on impact.



Ridges resulting from rapid cooling

Dramatic form | Rough | Some moldavites were still fluid when they were flung from the crater, and as a result tend to assume frothy, "splatter"-like forms such as this.



Pavilion faces upwards

Faceted moldavite | Cut | This particularly dark green moldavite gemstone has been faceted in a freeform shape, viewed here from the pavilion.



Crown facets

Excellent clarity | Cut | Certain moldavite gemstones are exceptionally transparent, such as this highly refractive, brilliant-cut example.

Moldavite takes its name from the town of Moldauthein in the Czech Republic



Horse head | Carved | The carver of this moldavite shaped in the form of a horse's head has left the natural surface of the mineral as the horse's mane.